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Large-Scale and Highly Reliable Hopfield Neural Networks Using Vertical NAND Flash Memory for the In-Memory Associative Computing

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Large-scale and highly reliable Hopfield neural networks (HNNs) for associative computing are implemented by vertical NAND (VNAND) flash memory for the first time. Fueled by the vertically stackable word-line (WL) stacks and sequential page-by-page read-out operations of a VNAND architecture, the HNN with 2^{10}-nodes capable of asynchronous updates can be implemented in a single block, featuring remarkable proficiency in restoring 2^{10}-bit (1 Kb) images corrupted by various types of noise. Utilizing asynchronous update scenarios, the VNAND HNN achieve a 13% increase in restoration accuracy and a 60.4% improvement in update speed compared with synchronous updates. Additionally, the influence of fabrication variation on accuracy degradation is analyzed. It is observed that the weights fine-tuned by the incremental step pulse programming achieve <4% degradation rate. Furthermore, association performance is evaluated using restoration test from the untrained MNIST images, demonstrating a 98.7% association accuracy with the threshold (θ) modulation method for optimized asynchronous updates. Finally, thanks to the extremely low bit-cost of a VNAND structure, the VNAND HNN occupies ≈117× smaller area than a 1T1R resistive random-access memory HNN under the same technology node.

1. Introduction

Recent advancements in artificial intelligence (AI) have significantly expanded the scope of machine capabilities, enabling the execution of human-like cognitive tasks such as pattern recognition, natural language generation, and associative computing.[1–3] These tasks are addressed with a variety of neural network (NN) architectures, which are specialized for each data type and problem domain, including convolutional NNs (CNNs), transformer-based large language models (LLMs), and recurrent NNs (RNNs).[4–6] Among them, the Hopfield NN (HNN), a single-layer RNN characterized by an energy-based computational framework and a fully interconnected neuron structure, has garnered particular attention for its applicability in associative computing and optimization problems.[7–9]

In HNNs, synaptic weights (W_{ij}) are stored between neurons (neuron i and j) in a single layer after training and utilized in iterative update process for the input values (V_n) of each neuron, as shown in **Figure 1**a,b. Due to the structural characteristics of memory-based HNNs, the number of neurons (N) and the size of synaptic weight matrix (N^2) abruptly increase as the dimension of NNs becomes larger for complex neural applications. For instance, while only 10^2 neurons and 10^4 weights are needed for solving the traveling salesman problem (TSP) of 10 cities in which the city-to-city connections are represented by neurons, 10^4 neurons and 10^8 weights should be utilized to solve the TSP of 100 cities, necessitating excessive data storage.[10,11] Furthermore, as the weight matrix is repeatedly multiplied by the input vectors in update cycles (vector-matrix multiplication, VMM), HNNs are subjected to memory-intensive operations as the NN size scales up. Thus, large-scale HNNs based on the conventional von Neumann architectures face critical limitations, primarily stemming from excessive data movement between memory and processing units, thereby leading to higher power consumption, latency, and lower overall computational efficiency.[12]

To overcome these challenges, various pioneering neuromorphic architectures have been proposed, integrating computational functionalities directly within memory. Nonetheless, most of previous HNN hardware based on resistive random-access

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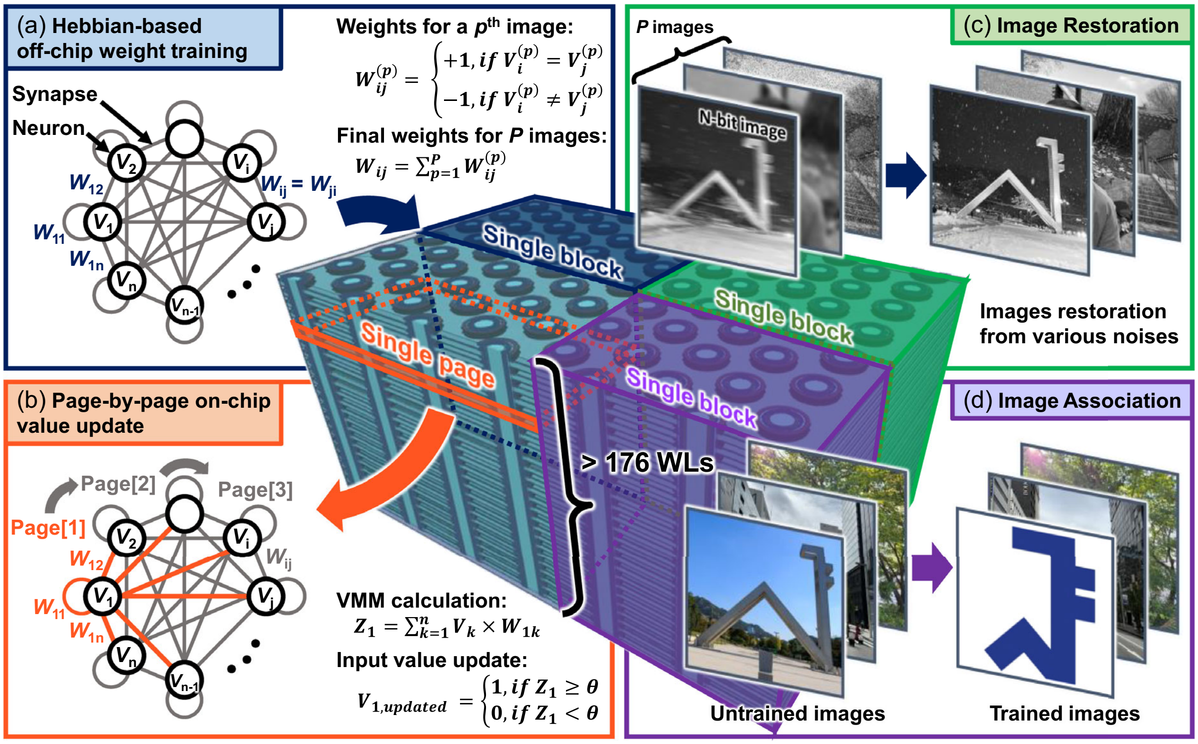

Figure 1. Conceptual view of the proposed VNAND HNN architecture. a) Weight training based on the Hebbian learning rule. The trained weight matrix is stored in a single VNAND block after the off-chip training. b) Page-by-page value update scheme of the VNAND HNN. Each value is updated by the result of the vector-matrix multiplication within the memory. c) Image restoration and d) image association are performed in a single VNAND block.

memory (RRAM) and static RAM (SRAM) have exhibited critical limitations in terms of scalability, integration density, and reliability. For example, RRAM-based HNNs generally suffer from severe device-to-device variation, restricting their size to relatively small values (≈100 neurons).[13–15] Conversely, SRAM-based HNNs require large chip area, severely limiting their feasibility for practical large-scale applications.[16,17]

In this work, a novel neuromorphic architecture based on vertical NAND (VNAND) flash memory is introduced, which is specifically tailored to achieve efficient and reliable large-scale HNN computations (>1000 neurons), as illustrated in Figure 1. Unlike conventional planar memory structures, the VNAND architecture efficiently integrates an N^2-weight matrix by allocating N synaptic weights within each memory page and vertically stacking multiple pages through word-lines (WLs), which significantly enhances the dimensionality of the network's energy surface.[11,18] Additionally, the intrinsic page-by-page operational mode of VNAND naturally aligns with the asynchronous update method, as each page independently stores the synaptic weights corresponding to a single neuron, as shown in Figure 1b. Thus, the VNAND HNN can efficiently perform asynchronous updates, making it highly suitable for complex associative computing while remaining fully compatible with conventional VNAND operational paradigms. Owing to the high integration capability of the vertical stack and the alignment with asynchronous updating, the proposed architecture supports 1024-bit (1 Kb) ($N = 1024$) image restoration and association tasks within a single memory block, as illustrated in Figure 1c,d.[19,20]

To demonstrate the performance advantages of the proposed VNAND HNN, extensive device simulations and experimental validations are conducted, particularly in terms of restoration accuracy, update speed, and device variation robustness. Our results confirm that the asynchronous update methods in the proposed architecture exhibit accurate and high-speed associative computing with robustness against device variations. Furthermore, we highlight the potential for enhanced image association accuracy by modulating the update threshold θ.

2. Results and Discussion

2.1. Device and Architecture Characteristics of the VNAND HNN

The proposed VNAND HNN architecture employs the Hebbian-based weight training method in which the synaptic weight W_{ij} is determined by the values of connected neurons (V_i and V_j, which are binarized to 1 and 0) as illustrated in Figure 1a. For each training image vector p, if two connected neurons share the same value ($V_i^{(p)} = V_j^{(p)}$), a weight of $+1$ is stored in $W_{ij}^{(p)}$; conversely, if they have different values ($V_i^{(p)} \neq V_j^{(p)}$), the weight of -1 is stored.[7] When P images are trained, the final synaptic weights (W_{ij}) are determined by summing the individual weight contributions from all the images. Thus, the final stored weights range from $-P$ to $+P$, reflecting the cumulative correlation or anticorrelation between neuron states across multiple training images. Exceptionally, the diagonal elements of the weight matrix (W_{ii}) are conventionally set to zero. The architecture is designed to operate as an associative memory using integer weights ranging from −15 to +15, implemented by the two-transistor-and-two-NAND-string (2T2S) configuration in which even and odd strings store the positive and negative components, respectively.[21] Thus, even when the number of training images P increases, the accumulated weights are quantized to the fixed range of −15 to +15 due to the reliable weight implementation of the memory cells. Finally, each VNAND page stores a unique set of weights associated with one neuron and its pairwise interactions, as each training dataset is composed of distinct value vectors.

To ensure that these multilevel weight states are physically realizable in the memory array, we investigated the electrical characteristics of VNAND cells under various programming conditions based on commercially fabricated 90-WL VNAND flash memory.[22,23] Specifically, 3D device simulations were conducted for the top four VNAND cells, calibrated referring to the experimental data of fabricated VNAND devices, as shown in **Figure 2**a. The simulation employed nonlocal tunneling and Shockley–Read–Hall recombination models, with material parameters derived from previous studies.[24–26] As shown in Figure 2b, 16 discrete synaptic weight levels (from level 15 to 0) are reliably implemented within the linear region of the I_{BL}–V_{WL} transfer characteristics using the incremental step pulse programming (ISPP).[27] Thus, the I_{BL} is changed according to the weight level at the constant WL read voltage. For better data retention and endurance, the memory cells were designed to operate under lower WL read voltage ($V_{\mathrm{WL,read}} = \approx 0\ \mathrm{V}$) and WL pass voltage ($V_{\mathrm{WL,pass}} = \approx 4\ \mathrm{V}$) than conventional multilevel flash memories.[28] Additionally, low bit-line (BL) read voltage ($V_{\mathrm{BL,read}} = 0.5\ \mathrm{V}$) reduces power consumption during repetitive network update cycles, contributing to the overall energy efficiency of the VNAND-based neuromorphic system.[29]

Building on the device-level characterization, we now describe how the proposed VNAND structure operates at the system level to perform associative computing. The proposed page-by-page value update cycles in a single VNAND HNN block are shown in **Figure 3**. Through an update cycle, the string-select-line (SSL) input voltages (1 and 0), multiplied by the weights stored in the selected VNAND page, determine the string current. For the page-by-page update operation, a read voltage (V_{read}) is applied to the WLs of the sequentially selected pages, while a pass voltage (V_{pass}) is applied to the other WLs, as shown in Figure 3b. Subsequently, the current sums from the even and odd BLs are compared using differential sense amplifiers (DSAs) to decide the input value update corresponding to target page, which is expressed as

$$V_{i,\mathrm{updated}} = \mathrm{sign}\left(\sum_{k=1}^{n} V_k \times W_{ik} - \theta\right) \tag{1}$$

where $V_{i,\mathrm{updated}}$ is the updated value of the neuron associated with the selected ith page, and the weights W_{ik} are stored across the page. The threshold θ serves as a bias term to modulate the neuron sensitivity to the aggregated weighted input, which is typically adjusted within the DSA circuits. This update rule is mathematically equivalent to the asynchronous update scheme of HNNs.[7] Thus, the page-by-page operation of VNAND inherently realizes asynchronous HNN dynamics with high structural compatibility.

To physically implement an HNN with N neurons in the architecture, a single VNAND block should consist of N-WL and $2N$-cells/page to represent both positive and negative weights. However, while recent VNAND technologies support over 16 k cells/page, the maximum number of WL stacks is limited to ≈500, creating a structural mismatch between vertical and horizontal scalability.[30] To resolve this limitation, we propose subdividing the large VNAND block into multiple sub-blocks, each capable of supporting a portion of the full neuron array, even as the memory area increases, as illustrated in **Figure 4**. For instance, rather than constructing a single block with 1 k-WL × 2 k-cells/page for 1 kb ($N = 1000$) image processing,

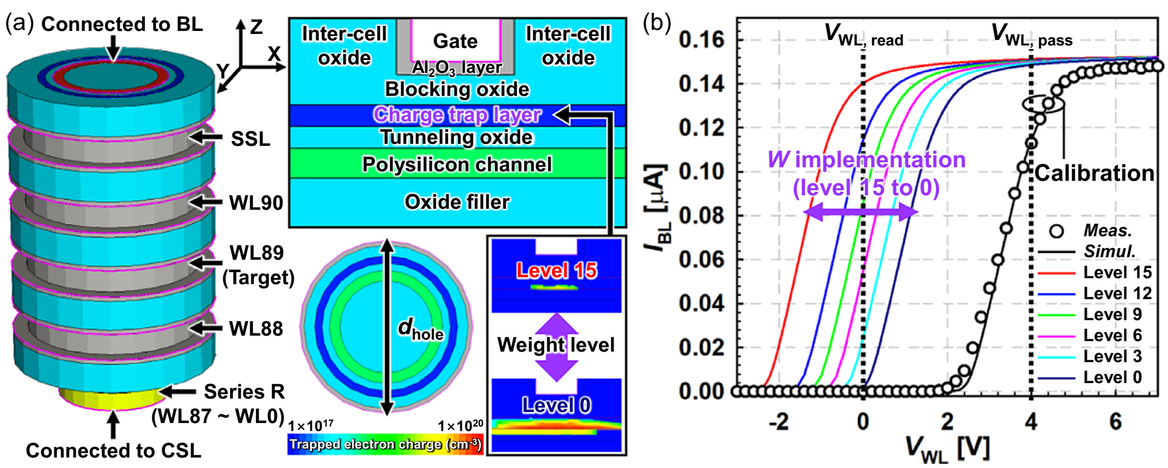

Figure 2. a) Simulated structure of top four-cell VNAND string. The remaining bottom cells (WL87 ≈ WL0) are modeled collectively as an equivalent series resistor. The vertical (y–z plane) and horizontal (z–x plane) cross-sectional view is also illustrated. b) I_{BL}–V_{WL} characteristics with various weight levels and the experimental data (90-WL VNAND device) for calibration. 16 levels of weight (15–0) are stored by modulating the trapped electron charge in the charge trap layer, with only the 6 levels of weight displayed in the I_{BL}–V_{WL} curve.

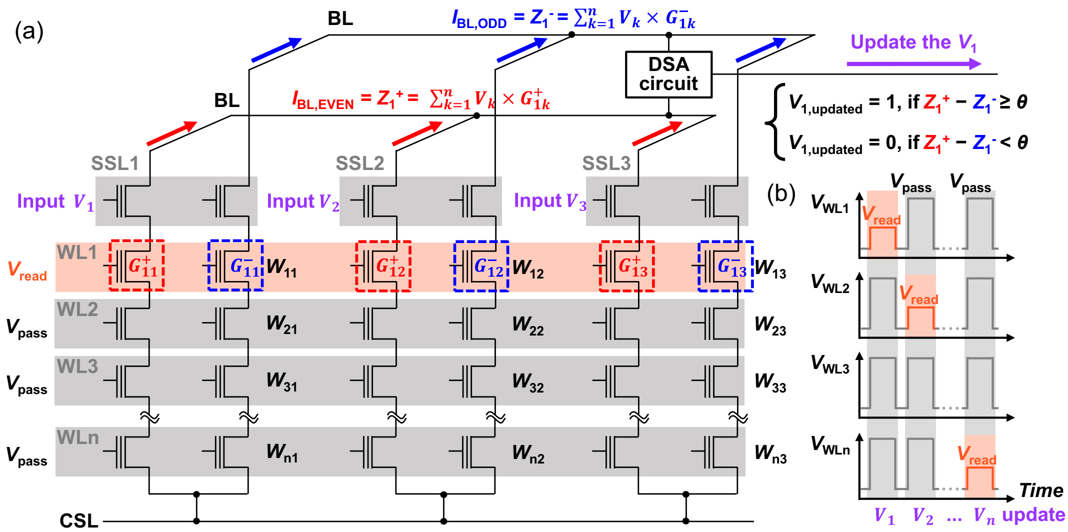

Figure 3. a) Input value update scheme of a single VNAND HNN page by performing VMM. Two VNAND cells are utilized as a pair for integer weight implementation. The input value V_i is updated at ith update step (same as a read scheme of ith WL page) and becomes an updated input at the next update step. b) Bias scheme of the proposed page-by-page update scheme.

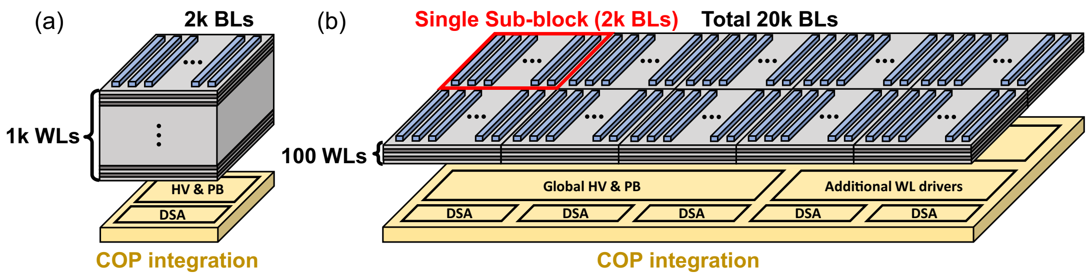

Figure 4. The architecture of VNAND HNNs a) in a single block with 1 k-WL × 2 k-cells/page and b) in 10 sub-blocks with 100-WL × 20 k-cells/page. The COP integration is utilized in both configurations. The sub-block partitioning is introduced for the practical hardware implementation of the commercially available VNAND flash memory.

a block with 100-WL × 20 k-cells/page partitioned into 10 sub-blocks is implemented, each with 2 k cells/page configuration. Thus, each sub-block independently performs operations corresponding to subsets of WLs (e.g., WL0–WL99, WL100–WL199, etc.), enabling effective realization of large-scale HNNs within the constraints of the number of WL stacks in conventional VNAND devices. Owing to the page-wise and sequential update cycle, sub-blocks share the peripheral circuits, such as the high-voltage (HV) charge-pump, sense amplifier (SA), and page buffer (PB), and only the local WL selection drivers and decoders are added per sub-block. Moreover, the cell-over-periphery (COP) integration, most of the peripheral circuits reside beneath the array, further mitigating any area penalty from sub-blocking.[31,32] The overhead in delay and energy is also negligible due to the μs-scale page-read transfer dominating update latency, while sub-block selection and decoding are ns-scale, and the page-wise VMM energy is dominated by BL conduction and SA/page-buffer terms, with incremental WL-selection switching remaining secondary.[33] In this work, the restoration and association of 1 Kb images are conducted in a single VNAND HNN block with 16 k cells/page and 176 WL stacks, which is divided into 6 sub-blocks.[34]

Given the hardware structure, the neuron update cycle should also be carefully configured to align with the physical memory access scheme. The update cycle can operate in two modes, depending on the configuration of the peripheral circuits: the synchronous update and the asynchronous update.[7] Synchronous update calculates VMM results for all neurons and subsequently updates their states collectively, ensuring uniform state transitions across the entire network, as shown in **Figure 5**a. In contrast, the asynchronous update method depicted in Figure 5b sequentially updates each neuron's state immediately after its individual VMM computation, allowing subsequent neurons to utilize the most current states. This sequential approach is well-suited to the intrinsic page-by-page operation of VNAND and reduces the total number of update operations, which results in a substantial reduction in energy consumption and circuit complexity.[35]

Figure 5c illustrates the network energy evolution under these two scenarios, highlighting that the synchronous update does not

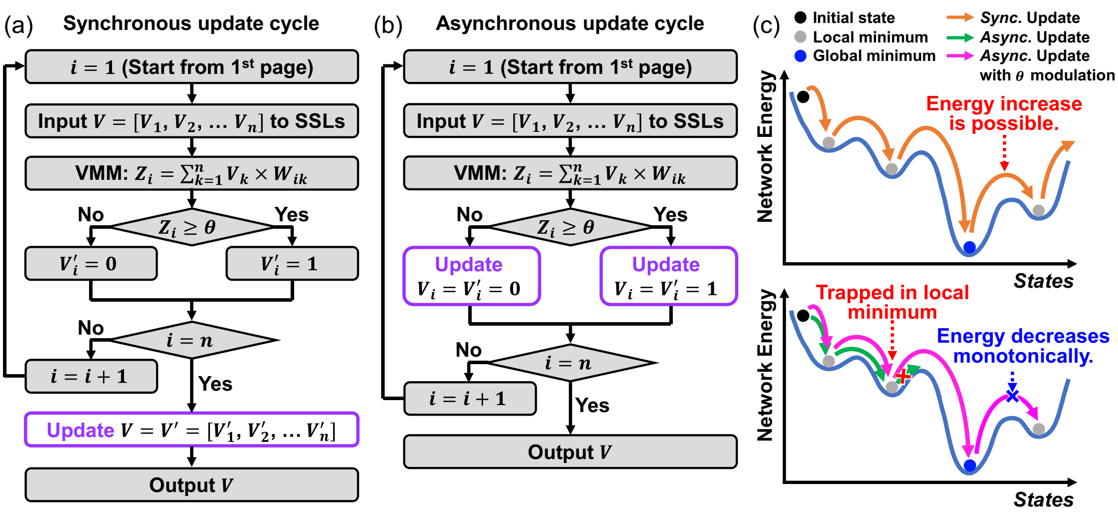

Figure 5. Flow chart of a) the synchronous-update cycle and b) the asynchronous update cycle in the VNAND HNN. c) State evolution on the network energy surface under two update scenarios. States denote the value vectors after the update cycle, with a fixed weight matrix.

guarantee a monotonic decrease in network energy, due to the concurrent updating of the neurons from the same previous states. The nonmonotonic update of network energy may result in nonconvergent behavior as the training data increases, such as oscillations where two states with similar energy levels repeatedly alternate.[7] Conversely, asynchronous update yields a consistent reduction of network energy and stable convergence, because each update aligns a single neuron with its local field. However, this approach could trap the system in local minimum, preventing it from finding globally optimal solutions. Thus, the advanced update methods are applied to the asynchronous update, such as simulated annealing and threshold (θ) modulation.[36]

2.2. Associative Computing of the VNAND HNN

Associative computing is a computation paradigm inspired by the function of the human brain, which stores and retrieves information based on partial input or pattern similarity, rather than explicit memory addresses. Leveraging this principle, the proposed VNAND HNN architecture performs image restoration under an asynchronous update scenario, where corrupted images are reconstructed based on learned associations, even in the presence of various noise types: salt and pepper, vertical, and horizontal noises (**Figure 6**a). The various 1 Kb images are trained following Hebbian learning rule-based weight matrix calculations, and the weights are quantized to a range (−15–15) optimized for VNAND HNNs, as shown in Figure 6b.

The effectiveness of the proposed architecture in the restoration process is presented in Figure 6c, which presents the restoration accuracy as a function of noise corruption levels. The results show that images corrupted by <40% vertical and horizontal noise are restored with >80% accuracy across different noise levels. The accuracy decreases as the number of images increases, but this can be mitigated by enhancing the resolution of the weights, corresponding to the storage capacity of the architecture. It is observed that the images corrupted by the salt and pepper noise show abrupt restoration accuracy degradation with >50% corruption rate due to its high randomness. Furthermore, when the asynchronous input updates are applied to 30 test images, network energy decreases steadily along the update counts for all the images, while avoiding oscillation and trapping in a local minimum, as shown in Figure 6d.

To further evaluate the efficiency of the update strategy, the restoration performance of asynchronous and synchronous-update methods is compared in terms of accuracy and convergence speed. Even as the number of stored patterns increases, both update methods still achieve perfect retrieval in the best cases. However, as shown in **Figure 7**a, the asynchronous update yields higher average and minimum restoration accuracy than the synchronous update when the number of images is >10. It is because synchronous updates lead to nonmonotonic changes in the network energy surface, which may cause oscillatory behaviors or prevent convergence in complex networks overloaded with patterns.[7,8] In addition to accuracy, this instability also affects convergence efficiency. As shown in Figure 7b, asynchronous updates require fewer update cycles on average and exhibit more consistent convergence behavior across different noise levels and the number of images.

While asynchronous updates offer improved accuracy and convergence stability, their robustness under hardware variation conditions must also be considered. In particular, the precision of weight storage in VNAND cells is affected by device-level characteristics such as threshold voltage (V_{th}) variation and the slope of the linear region ($g_{m,max}$) in the I_{BL}–V_{WL} curves. In the proposed VNAND HNN, weights are stored in the linear region of the I_{BL}–V_{WL} curve, by adjusting the V_{th}. However, when V_{th} variation becomes severe, the network may converge to incorrect outputs, and the accuracy degradation is further exacerbated if the $g_{m,max}$ is too steep. Thus, achieving gradual $g_{m,max}$, and minimizing V_{th} variation are essential for reliable and high-resolution weight implementation. Their dependance on the device parameter and fabrication variation is investigated in **Figure 8**. Figure 8a shows measured I_{BL}–V_{WL} curves with the variation of the channel hole diameter (d_{hole}) during one-shot

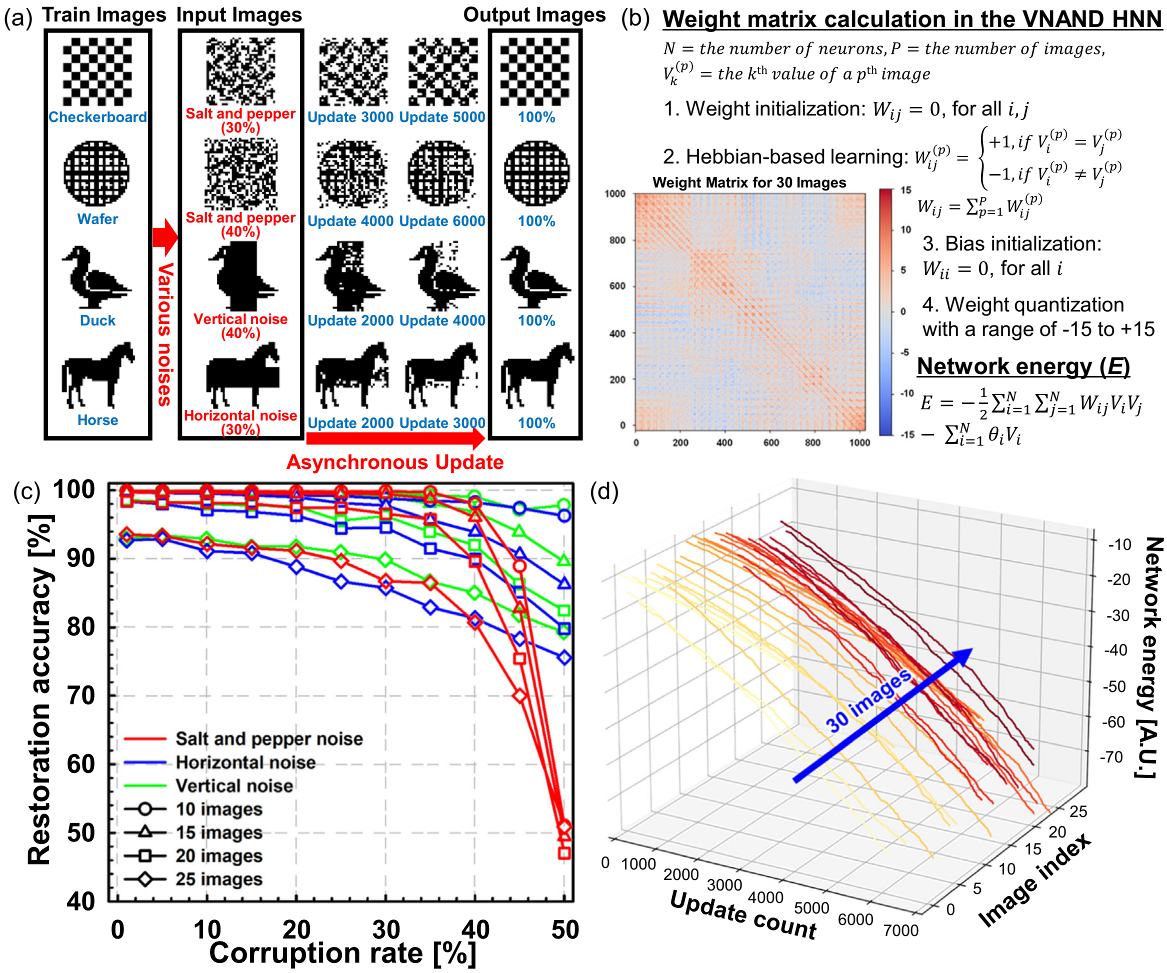

Figure 6. a) Image restoration process from various noise sources. b) Calculation of weight matrix and network energy in the VNAND HNN. c) Restoration accuracy with varying the corruption rate. d) Network energy trend over the number of asynchronous value updates.

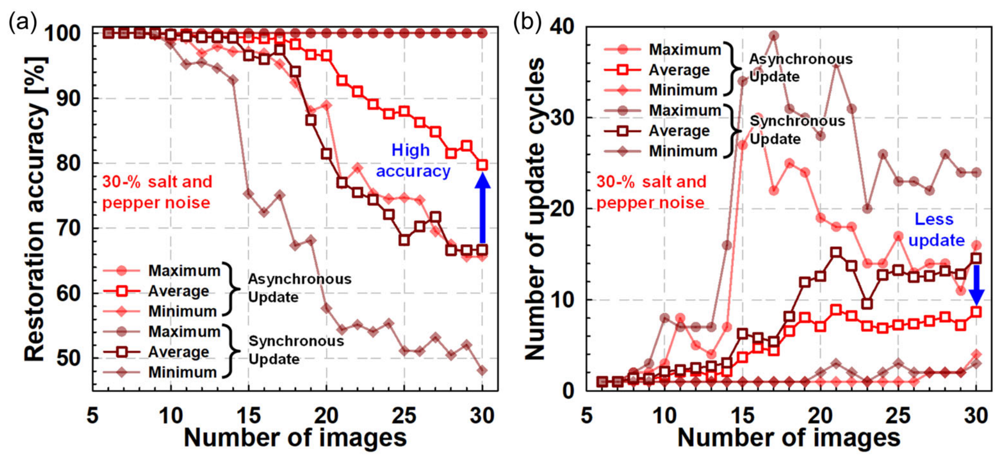

Figure 7. a) Restoration accuracy and b) the number of update cycles for 30% salt and pepper noise under the asynchronous and synchronous update. The number of update cycles indicates the time required for the network to converge for each image to be restored.

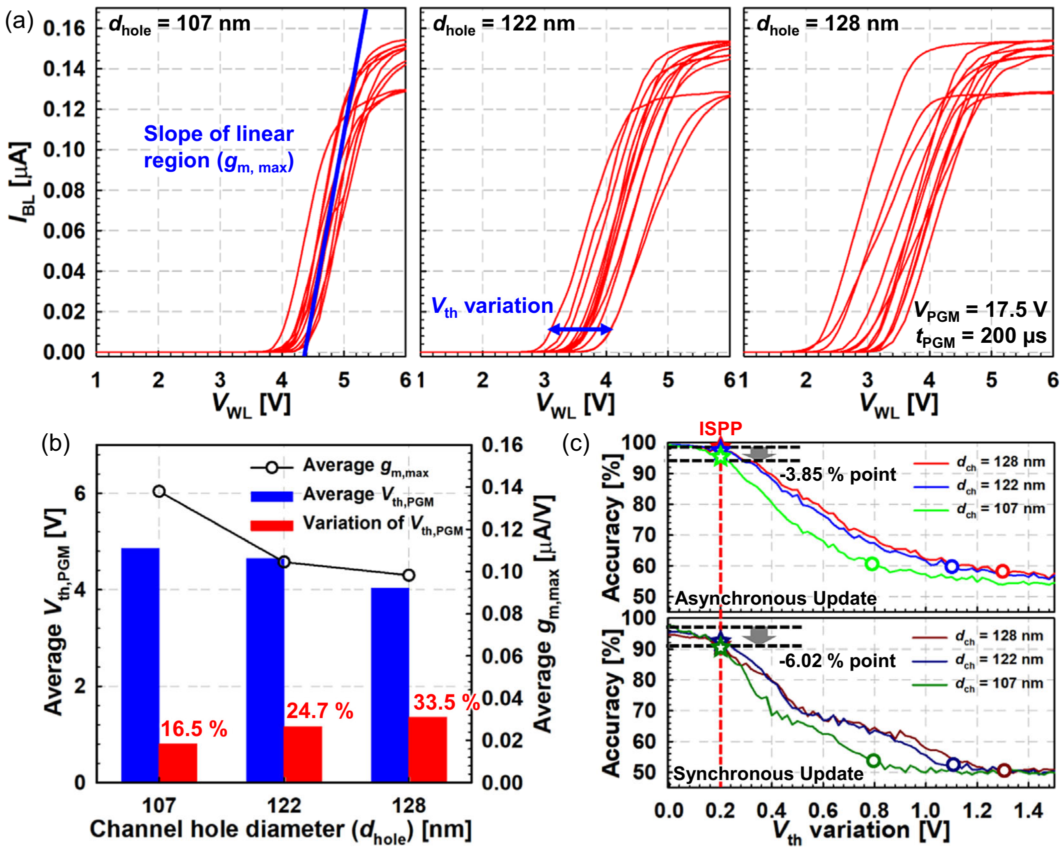

Figure 8. a) Measured I_{BL}–V_{WL} curves of the 90-WL VNAND string after one-shot program with the variation of d_{hole}. b) Comparison of $g_{m,\ max}$, V_{th}, and V_{th} variation of the VNAND cells. The ratio of variation to V_{th} is also specified. c) Accuracy degradation as a function of the V_{th} variation under asynchronous and synchronous-update scenarios. Each circle and star symbol indicates the accuracy after one-shot program and ISPP. Restoration of 15 images from 30% salt and pepper noise is conducted to compare both scenarios with the same initial accuracy.

program condition. As d_{hole} decreases due to scaling, gate controllability improves, leading to a steeper $g_{m,max}$, which in turn narrows the weight-implementable region. Nonetheless, this can be mitigated through device-level optimization, such as adjusting the thickness of the channel and oxide layers.[37]

In general, as the critical dimension (CD) shrinks, devices become more sensitive to local nonuniformities, such as line-edge roughness and channel-hole geometry irregularities, leading to larger V_{th} variation. However, interestingly, the V_{th} variation becomes slightly alleviated even though the charge capacity of charge-trapping layer decreases due to the d_{hole} downscaling. This is attributed to enhanced gate controllability, which reduces the sensitivity of device characteristics to physical profile variations.[38,39] Figure 8c shows the practical accuracy degradation, which is considered with both the $g_{m,max}$, and V_{th} variation. Even if the $g_{m,max}$ is lower with larger d_{hole}, it is observed that large V_{th} variation after one-shot program induces severe accuracy degradation. However, the accuracy degradation is alleviated below 4% with ISPP, which achieves a small and uniform V_{th} variation (≈0.2 V) condition.[21] In addition, asynchronous updates exhibit superior robustness to device-level variation compared to synchronous updates, under this fine-tuned weight configuration. Because neuron states are updated sequentially, the impact of localized weight variation can be progressively compensated in later updates, preventing the error from propagating across the network. In contrast, synchronous updates reflect all variations at once, potentially amplifying inaccuracies across neurons. This architectural and temporal decoupling enables asynchronous updates to maintain higher inference accuracy under varying conditions.

These architectural and device-level advantages highlight the superiority of the proposed VNAND HNN over RRAM-based implementations, which often suffer from greater device-to-device variation and long-term retention instability.[40,41] To further validate the practical scalability of the proposed architecture, the associative computing performance of the VNAND HNN and conventional RRAM HNNs is compared in terms of memory capacity, as shown in **Figure 9**a. Even with analog weight implementation in RRAM HNNs, reliable restoration accuracy could not be achieved beyond 15 images, primarily due to limited memory density and structural constraints of planar memory arrays. In contrast, the VNAND HNN demonstrates robust performance across over 50 images, owing to its high memory density. Figure 9b quantifies this advantage by comparing the chip area as a function of the number of neurons processed simultaneously. A VNAND HNN block with 16 k cells/page and 176 WL stacks implements ≈1760× and ≈117× higher memory density than a SRAM one and a RRAM one, assuming the same design

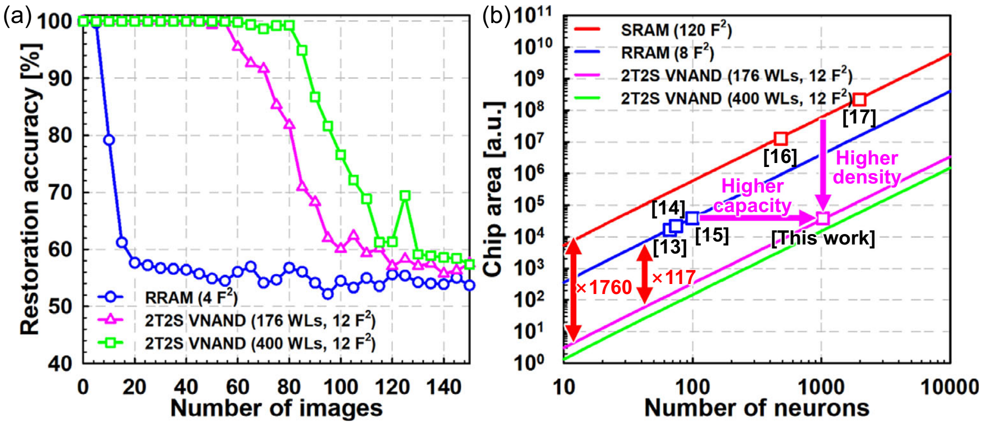

Figure 9. a) Average restoration accuracy of RRAM HNN and proposed VNAND HNN for 10% salt and pepper noise under the same chip area. The number of neurons (N) is set to 100 for RRAM, 1000 for 176-WL VNAND, and 1300 for 400-WL VNAND to match the respective architectural capacities. b) Chip area of the SRAM, RRAM, and VNAND-based HNNs as a function of the number of neurons computed simultaneously. The memory cell area (F^2) is compared under the identical technology node ($F = 50$).

and process node.[42] Thus, ≈10× higher neuron capacity than an RRAM one is achieved in VNAND HNN at the same chip area. The capability of the proposed VNAND HNN can be further enhanced by employing additional memory blocks and increasing the number of WL stacks.[30] The power consumption and update speed of VMM are also analyzed with the reliability metrics for practical in-memory computing applications, as summarized in **Table 1**. To compute power and latency under the identical conditions, a 1024 × 1024 VMM operation with a synchronous-update scenario is assumed.[43,44] Due to the page-wise operation of VNAND, the number of WL is multiplied to obtain the block-level energy and time consumption, which

Table 1. Benchmark of the neuromorphic architecture for HNNs.

	Device variation (σ_W/μ_W)	Accuracy degradation	Retention at 125 °C	Energy/VMM [μJ]	Update speed ($N = 1024$) Sync.	Update speed ($N = 1024$) Async.	Effective area/synapse ($F = 50$ nm)
SRAM HNN	Excellent	Negligible	Volatile	5.14[43]	Fast[43] (40 μs)	Slow[43] (40.9 ms)	250 000 (nm^2)
RRAM HNN	Poor[40] (25–30%)	High (>30%)	Good[41] (200 min)	7.06[43]a)	Fast[43]a) (25 μs)	Fast[43]a) (25.6 ms)	10 000 (nm^2)
VNAND HNN (176 WLs)	Good[18] (1–3%)	Low (≈5%)	Good[28] (360 min)	7.21a)	Slow a) (25.6 ms)	Fast a) (25.6 ms)	170 (nm^2)

a)Same read speed $t_{read} = 25$ us is assumed.

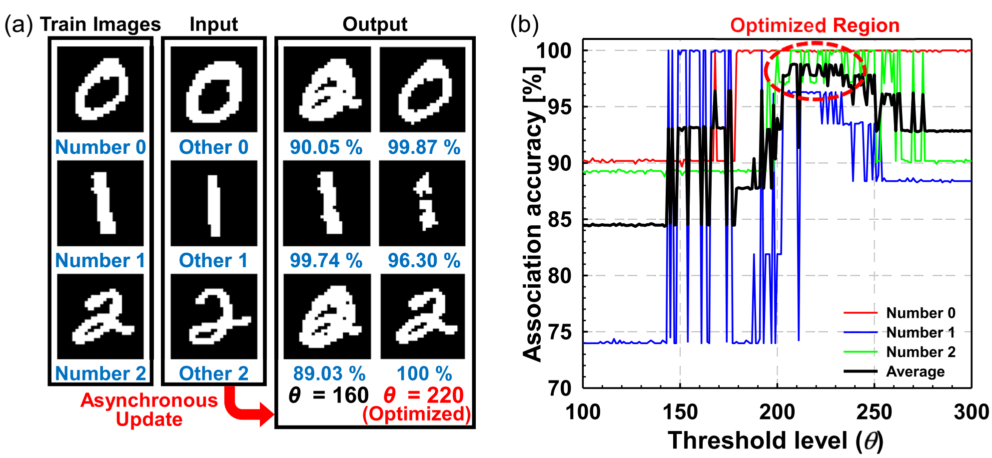

Figure 10. a) Image association process of VNAND HNNs for untrained MNIST images. b) Image association accuracy according to the threshold (θ) modulation.

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degrades the performance. However, if an asynchronous update scheme is adopted, the effective throughput of RRAM and SRAM would also scale with the number of neurons.

Beyond the image restoration, the VNAND HNN also demonstrates image association capabilities, identifying the most relevant images from untrained inputs based on pattern similarity, as shown in **Figure 10**a. This task expands the functionality of VNAND HNNs beyond exact recall, highlighting their potential for more general associative computing. However, compared to image restoration, which involves recovering a previously stored pattern from a noisy input, image association is a more challenging task, as it requires the network to map unseen inputs to the closest attractor in its energy landscape. While restoration typically benefits from smooth convergence toward known energy minima, association must navigate a more ambiguous energy surface, increasing the likelihood of convergence to incorrect or spurious attractors. To overcome these limitations, a threshold modulation method is introduced for the update cycle to circumvent local minimum.[36] Threshold modulation within the network is feasible in two methods. The first approach embeds the threshold value θ directly into the memory cells by assigning a nonzero value to the diagonal elements of the weight matrix W_{ii}, which represent self-feedback to each neuron. It contrasts with the typical Hopfield formulation where $W_{ii} = 0$ and allows the threshold to be locally encoded within the memory array. The second method adjusts the threshold externally using the DSA circuits, which evaluate the VMM outputs during the update process. This circuit-level approach enables flexible and real-time modulation of threshold values without altering the stored weights. Figure 10b demonstrates that modulating the threshold has a significant impact on the network's performance. By sweeping across various threshold levels, a specific value was found to yield the highest performance, indicating that the optimal threshold is network-dependent. This result suggests that threshold modulation is an effective strategy to improve associative computing reliability in VNAND HNNs.

3. Conclusion

In this work, we proposed a novel neuromorphic hardware architecture that implements large-scale HNNs using VNAND flash memory. Leveraging the high-density and high-reliability characteristics of VNAND, the proposed VNAND HNN supports over 1 k neurons within a single memory block, enabling in-memory associative computing with significantly reduced area compared to conventional SRAM- or RRAM-based implementations. We demonstrated that the HNN implemented in a single block of VNAND can robustly restore corrupted 1024-bit images and successfully perform image association tasks, achieving up to 98.7% accuracy on untrained MNIST inputs. The inherent page-wise structure of VNAND facilitates asynchronous update operations, which not only improve convergence stability by ensuring monotonic energy descent but also enhance accuracy and reduce update cycles compared to synchronous methods. Overall, this work establishes VNAND as a promising platform for scalable, robust, and energy-efficient associative computing, bridging the gap between memory technology and brain-inspired neural network hardware.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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associative computing, Hopfield neural networks, neuromorphic hardware, vertical NAND flash memory

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